

Efthymios Costa

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🌐 GitHub Profile
🌐 Personal Website

EDUCATION

PhD in Statistics and Machine Learning (StatML)

Oct 2021 - Sep 2025

Imperial College London, Department of Mathematics

Supervisors: Dr Ioanna Papatsouma & Professor Alastair Young

Thesis title: *Novel unsupervised techniques for mixed-type data*

Visiting Doctoral Student

Jan - Mar 2024

University of Bologna, Department of Statistical Sciences "Paolo Fortunati"

Visiting Supervisor: Professor Christian Martin Hennig

MSc Statistics (Data Science)

Sep 2020 - Sep 2021

Imperial College London, Department of Mathematics

Distinction (80.89/100)

Thesis: "A benchmarking study of distance-based clustering methods for mixed-type data"

BSc Mathematics

Sep 2017 - Jun 2020

Imperial College London, Department of Mathematics

First Class Honours (78.21/100)

EMPLOYMENT

Research Associate with Teaching Duties in Statistics

Oct 2025 - Present

Imperial College London, Department of Mathematics

London, United Kingdom

Postdoc & Fellows Representative in the Department of Mathematics

RESEARCH INTERESTS

Unsupervised learning: Cluster analysis, outlier detection, dimensionality reduction.

Mixed-type data: Combination of continuous and discrete variables in data.

Robust statistics: Robustness guarantees for machine learning methodologies.

PUBLICATIONS & PREPRINTS

Efthymios Costa, John R.J. Thompson (2026). "Spectrally Tuned Bandwidth Selection for Kernel Fuzzy Relational Clustering". *arXiv preprint arXiv:2607.03117*.

Efthymios Costa, Christian Hennig (2026). "A unified approach to outlier identification for mixed-type data". *arXiv preprint arXiv:2606.26324*.

Efthymios Costa, Ioanna Papatsouma, Angelos Markos (2026). "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data". *Pattern Recognition* 179, pp. 113580

Efthymios Costa, Ioanna Papatsouma, Angelos Markos (2026). "Sparse clustering via the Deterministic Information Bottleneck algorithm". *arXiv preprint arXiv:2601.20628*.

Efthymios Costa, Ioanna Papatsouma (2025). "A novel framework for quantifying nominal outlyingness". In: *Statistics and Computing* 36, pp. 41–58.

Efthymios Costa, Ioanna Papatsouma (2025) Contribution to the discussion of Dümbgen and Davies (2025) "Connecting Model-Based and Model-Free Approaches to Linear Least Squares Regression". In: *Statistica* 84, pp. 107–108.

Efthymios Costa, Ioanna Papatsouma, Angelos Markos (2022). "Benchmarking distance-based partitioning methods for mixed-type data". In: *Advances in Data Analysis and Classification* 17, pp. 701–724.

FUNDING & AWARDS

2026 Distinguished Dissertation Award

The Classification Society

Poster Competition 1st Place & People's Choice Award

Faculty of Natural Sciences Research Showcase 2025, Imperial College London

Poster Competition People's Choice Award

Mathematics PhD Symposium 2025, Imperial College London

Best Postgraduate/Doctoral Paper Award

6th Annual Conference of the Cyprus Statistical Society, 2025

Faculty of Natural Sciences Prize for Excellence in Teaching

Highly commended nomination for excellence in teaching and learning, 2025, Imperial College London

MSc in Statistics Imperial College London Ambassador

Honorary title in recognition of outstanding contributions to the running and delivery of the programme

Poster Competition Runner-up & People's Choice Award

Mathematics PhD Symposium 2022, Imperial College London

PhD Studentship

EPSRC Centre for Doctoral Training (CDT) in Statistics and Machine Learning (StatML)

Warner Prize

Prize awarded for research project development in the MSc in Statistics at Imperial College London - Class of 2021

Outstanding Academic Representation Network Team Award

Student volunteer award - Imperial College Union Student Choice Awards 2020

QUALIFICATIONS

Associate Fellow of Advance HE (AFHEA)

Awarded on February 2024, Imperial College London

CONFERENCES, WORKSHOPS & SEMINARS

19th Conference of the International Federation of Classification Societies 14 - 16 Jul 2026

Presentation title: "Sparse clustering via the Deterministic Information Bottleneck algorithm" Invited Speaker

2026 Classification Society Annual Meeting 16 - 18 Jun 2026

Presentation Title: "Novel unsupervised techniques for mixed-type data" Distinguished Dissertation Award Address

19th International Conference on Computational and Methodological Statistics 13 - 15 Dec 2025

Session Title: "Clustering of heterogeneous data" Session Organiser/Chair

Joint meeting of the Italian and Dutch/Flemish Classification Societies (CLADAG-VOC 2025) 8 - 10 Sep 2025

Presentation title: "Cluster Analysis from an Information-Theoretic Viewpoint" Invited Speaker

International Conference on Robust Statistics (ICORS 2025) 19 - 23 May 2025

Presentation title: "A novel framework for quantifying nominal outlyingness" Invited Speaker

18th International Conference on Computational and Methodological Statistics 14 - 16 Dec 2024

Presentation title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data" Presenter

RSS Emerging Applications Section Meeting "Network and Clustering Analysis for Emerging Applications" 13 Dec 2024

Presentation title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data" Invited Speaker

UKRI EPSRC Mathematical Sciences Team visit at Imperial College London 9 Oct 2024

Poster title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data" Presenter

18th Conference of the International Federation of Classification Societies 15 - 19 Jul 2024

Presentation title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data" Presenter

Imperial Mathematics PhD Symposium 12 Jun 2024

Poster title: "A Novel Approach to Outlier Detection for Mixed-Type Data" Presenter

Department of Statistical Sciences "Paolo Fortunati", University of Bologna - Weekly Statistics Seminars 1 Feb 2024

Presentation title: "Outlier detection for mixed-type data: A novel approach" Presenter

16th International Conference on Computational and Methodological Statistics 16 - 18 Dec 2023

Poster title: "A novel approach to outlier detection for mixed-type data" Presenter

Amazon Development Center Germany GmbH 27 - 28 Mar 2023

Advanced training on Statistical Machine Learning and Computing for big-data analysis Attendee

Poster title: "Benchmarking distance-based partitioning methods for mixed-type data" Presenter

Faculty of Natural Sciences Research Showcase 2022

29 Sep 2022

Presentation title: "Clustering mixed-type data: Which method to choose?"

Presenter

Bocconi Summer School in Advanced Statistics & Probability

11 - 22 Jul 2022

Topic: Random Structures and Combinatorial Statistics

Attendee

Imperial Mathematics PhD Symposium

16 Jun 2022

Poster title: "A full factorial benchmarking study of non-parametric partitioning methods for mixed-type data"

Presenter

TEACHING EXPERIENCE

Module Lead

MATH40008 - Individual Research Project - BSc/MSci Mathematics Year 1 (Summer 2025-2026)

BSc/MSci Mathematics Final Year Project Supervisor

Imperial College London, Department of Mathematics (Academic Year 2025-2026)

MSc Statistics Dissertation Supervisor

Imperial College London, Department of Mathematics (Academic Year 2025-2026)

Senior Graduate Teaching Assistant

Probability and Statistics - BSc/MSci Mathematics Year 1 (Autumn 2024)

Probability for Statistics - BSc/MSci Mathematics Year 2 (Autumn 2022)

Imperial College London, Department of Mathematics

Statistics - MEng Mechanical Engineering Year 3/4 (Autumn 2023)

Imperial College London, Department of Mechanical Engineering

Graduate Teaching Assistant

Data Science - MSc Statistics (Spring 2022)

Computational Statistics - MSc Statistics (Autumn 2022, 2023 & 2024)

Unsupervised Learning - MSc Machine Learning and Data Science (Autumn 2022)

Deep Learning, Supervised Learning - MSc Machine Learning and Data Science (Spring 2023)

Machine Learning - MSc Statistics (Spring 2023)

Introduction to Statistical Learning, Statistical Modelling II - BSc/MSci Mathematics Year 3/4 (Spring 2023)

Statistical Modelling I - BSc/MSci Mathematics Year 2 (Spring 2023)

Imperial College London, Department of Mathematics

OUTREACH

Mentor for Imperial College London Mathematics School Research Project (2024 - 2025)

EDITORIAL & PEER-REVIEW SERVICE

Reviewer

Advances in Data Analysis and Classification

International Statistical Review

Journal of Classification

Mathematics

MEMBERSHIPS

Fellow of the Royal Statistics Society (RSS)

Member of the British Data Science Society (BDSS)

Member of the Cyprus Statistics Society (CSS)

SKILLS

Programming/IT: R (Advanced/Preferred), Python (Intermediate), C++ (Basic), MATLAB (Basic), $\text{\LaTeX}$, HPC projects on compute clusters, Office 365

Languages: Greek (native), English (fluent), French (proficient)